How accurate LIGHT v0_5 is

The basic network of LIGHT v0_5, version 0.1.5, on 11 October 2026: v0_4 with the 5s $\rightarrow$ 5p electron excitation on the cross sections Jung et al. measured, and a krypton-buffered e-beam laser held out, with its energy ladder and its checks

Bottom line. LIGHT v0_5 reproduces the fitted KrF decays [1, 2] within 6.1 % rms (2.3 to 6.1 %) and the decays no fit used [4] within 11.7 % rms, as v0_2 to v0_4 do. At the Tahoma laser [5] it gives the small-signal gain to -6 % (5.99 against the measured 6.39 %/cm; v0_4 -7 %), the absorption to -25 %, the net gain to -2 %, the saturation intensity to +4 % and the formation efficiency to -3 %. No fit used a laser's numbers and no rate was set by one.

What v0_5 changes is one rate: the electron pumping of the 5p rung, which v0_4's overview named as deciding the partition, runs on the cross sections Jung et al. measured out of the metastable levels of Kr [12, 13] in place of a 1976 fit [10, 11]. The measurement confirms the fit: its Maxwellian rate is $\times 1.37$ the fit's at 1.16 eV (Tahoma's electrons), $\times 1.11$ at 3 eV and $\times 0.91$ at 5.8 eV, and the 248-nm share at the reference point moves from 5.79 to 5.7 % of E_{dep} (5.68 to 5.73 % under the kinetic EEDF).

Its own uncertainty, propagated from its sources (Section 1: 22 inputs at the two ends their sources allow, in quadrature), is ± 38 % on the 248-nm share and ± 33, ± 37 and ± 40 % on Tahoma's gain, absorption and net gain, the largest term everywhere but the absorption the trimer's $(T_g/300)^n$, a band with no measurement behind it. A second laser now bears on that term: Kleikamp et al.'s krypton-buffered KrF laser [17] peaks at 2 atm of Kr, and LIGHT, scored with the paper's own saturated balance, puts the optimum at 1.26 atm (-37 %; 1.02 to 1.48 over the pump's uncertainty), because at 2 atm it quenches the threshold inversion $\times 2.2$ as fast as the laser's balance allows. Nothing is retuned.

The largest errors: Hakuta et al.'s C-A decays [6], 19.3 and 34.1 % rms; Burnham and Searles' KrF(B) lifetime [7], -22.3 %; the VUV yield of pure krypton at low pump [8], +32 % (v0_3's run); Tahoma's absorption, -25 %, 72 % of it Kr_2F^* at the cross section Hakuta et al. measured [9]; and the Kr pressure optimum of Kleikamp et al.'s laser, -37 %, with its F_2 optimum (10 mbar) not reproduced at all.

LIGHT v0_5 is v0_4 [22] with one row changed and two data sets added. K76, e^- + Kr(5s) $\rightarrow$ e^- + Kr(5p), was Kannari's Table I fit $A\,T_e^{1/2}\exp(-C/T_e)$ to e-beam laser data; it is now the lumped cross section built from the ten measured excitations out of each metastable level (Section 2), under both closures, with the fit kept as a switch. The TALIF review of Gazeli et al. [14] supplies a measured 5p′ lifetime that cross-checks the rung's radiative rate. Kleikamp et al.'s laser [17] is a second device, held out. Everything else, the 5p rung, the optics, KrF(B,C) + F, the ion-ion limbs, the walls, the wave and the kinetic EEDF, is v0_4's, and v0_4's overview describes it. The 10 kinetic datasets have no electrons, so v0_5 scores them exactly as v0_2, and `make_overview_v0_5.py` refuses to write a number that differs. Table 1 lists every test.

1 Overall accuracy and its uncertainty

Against the data. Table 1 scores every test; Table 2 reads the same numbers as errors, the model's deviation from the measurement in percent of the measurement (rms over a set of points), and names the largest.

test	measured	v0_5	v0_4	role of the data
energy ledger closure		below 3.4×10^{-14} with walls, 2.4×10^{-14} with the wave	the same	bookkeeping
KrF(B) decay against Kr and F_2, Eden 1978		rms 2.3 and 3.2 %	the same	train
KrF(B,C) slow decay, Gadomski 1992 Figs. 1 and 2		rms 6.1 and 4.2 %	the same	train
C/B emission ratio at 300 K, Yu and Setser 1990	0.16 ± 0.02	-0.5 %	the same	train
KrF(B,C) decays, Xu 1993 Figs. 5 to 7		rms 11.7 %	the same	held out
decay constants Xu 1993 printed, 50 and 100 Torr Kr	5.0×10^{7} and 6.2×10^{7} s^{-1}	$\times0.91$ and $\times0.97$	the same	structure check
C-A decays, Hakuta 1990		rms 19.3 and 34.1 %	the same	held out, in conflict
τ_B, Burnham and Searles 1977	9.0 ± 0.6 ns	-22.3 %	the same	held out, in conflict
C/B ratio 230 K over 300 K, Yu and Setser	1.30 ± 0.06	-8.4 %	the same	held out
Tahoma small-signal gain g_0, kinetic EEDF	6.39 %/cm	5.99, $\times0.94$	5.91, $\times0.93$	device, score only
the same, v0_2's closure		5.93, $\times0.93$	5.86, $\times0.92$	device, score only
Tahoma absorption α_0, kinetic EEDF	1.11 %/cm	0.83, $\times0.75$	0.82, $\times0.74$	device, score only
Tahoma net gain $g_0 - \alpha_0$, kinetic EEDF	5.28 %/cm	5.16, $\times0.98$	5.09, $\times0.96$	device, score only
Tahoma loss in pure Kr, 257.3 nm	0.67 %/cm	0.24, $\times0.36$ (248-nm cross sections)	the same	rough check
Tahoma saturation intensity I_{sat}, v0_2's closure	1.4 ± 0.1 MW/cm^2	1.45, $\times1.04$	1.47, $\times1.05$	device, score only
Tahoma formation efficiency $g_0 I_{sat}/P_{in}$, v0_2's closure	0.27	0.261, $\times0.97$	0.260, $\times0.96$	device, score only
Kr pressure optimum of Kleikamp 1986's Kr/F_2 laser, F_2 10 mbar	2 atm	1.26 atm (1.02 to 1.48 over the pump's spread), the paper's saturated balance	not scored	device, held out
F_2 pressure optimum, Kr 2 atm	10 mbar	above 30 mbar (still rising)		device, held out
three-body KrF^* + Kr quenching their balance deduces, cm^6/s	4.6×10^{-31} (model-dependent)	1.02×10^{-30} at the window's 362 K		device, score only

Table 1: LIGHT v0_5 against the data, beside v0_4, every band at its default unless a row says otherwise. A percentage is the deviation from the data, signed for a single value and rms for a set. A $\times$ factor is the model over the measurement. "The same" is checked: the generator stops if v0_5's number differs from v0_2's. The Kleikamp rows are new (Section 6).

quantity	measured	v0_5	error
KrF(B,C) decays the fit used, Eden 1978 and Gadomski 1992 [1, 2]	four sets		within 6.1 % rms (2.3 to 6.1)
KrF(B,C) decays no fit used, Xu 1993 [4]	three figures		within 11.7 % rms
C/B emission ratio at 300 K [3]	0.16 ± 0.02		-0.5 %
its 230 K/300 K ratio [3]	1.30 ± 0.06		-8.4 %
decay constants, 50 and 100 Torr Kr [4]	5.0×10^7, 6.2×10^7 s^{-1}	$\times0.91$, $\times0.97$	-9, -3 %
Tahoma small-signal gain g_0 [5]	6.39 %/cm	5.99	-6 %
Tahoma absorption α_0	1.11 %/cm	0.83	-25 %
Tahoma net gain $g_0 - \alpha_0$	5.28 %/cm	5.16	-2 %
Tahoma saturation intensity	1.4 ± 0.1 MW/cm^2	1.45	+4 %
Tahoma formation efficiency $g_0 I_{sat}/P_{in}$	0.27	0.261	-3 %
Kleikamp 1986's Kr pressure optimum [17]	2 atm	1.26 atm	-37 %
the largest errors			
C-A decays, Hakuta 1990 [6] (in conflict with the fitted sets)	two figures		19.3 and 34.1 % rms
KrF(B) lifetime, Burnham and Searles 1977 [7] (in conflict)	9.0 ± 0.6 ns		-22.3 %
VUV yield of pure Kr at low pump, Morozov 2008 [8], v0_3's run	0.42 ± 0.05	0.55	+32 %
Kleikamp 1986's Kr pressure optimum	2 atm	1.26 atm	-37 %
Kleikamp 1986's F_2 pressure optimum	10 mbar	above 30	not reproduced
Tahoma absorption α_0	1.11 %/cm	0.83	-25 %

Table 2: The scores of Table 1 as errors. A single value's error is signed, model less measurement over the measurement; a set's is the rms over its points. Tahoma's are under the kinetic EEDF, the Kleikamp optimum under v0_2's closure; the rest have no electrons.

Propagated from the sources. As in v0_4's overview, each input is set to the two ends its source allows and the outputs recomputed (Table 3, under v0_2's closure): the 17 bands to their ranges (Table 6; k_5s5p is now a scale on the measured cross section, 0.5 to 1.4, where in v0_4 it ran from the fit to 0); the 3 constants the model uses at a measured value whose source quotes an error and that no band covers, $Kr_2^+ + e^-$ (±10 % [19]), $Kr^+ + 2Kr$ (±9 % [20]) and the Kr_2F^* lifetime (±9 % [4]); and the 2 cross sections with a quoted error analytically, σ_{se} ±12 % [21, 10] and $\sigma(Kr_2F^*)$ ±20 % [9]. Half the difference between an input's two ends is its term; the quadrature sum treats the terms as independent and each as one standard deviation, which a band's range is not, so it is indicative; the linear sums of the excursions are the ceiling on the spread, not a bound. Not propagated: the electron-impact cross sections, now including Jung et al.'s, whose ±35 to 40 % the band k_5s5p carries instead, and the fitted KrF(B,C) set, which the training decays pin at 6.1 % rms.

On the 248-nm share at the reference point (5.68 % of E_{dep}) the quadrature sum is ±2.17 points (±38 %; v0_4 ±39 %), the excursions sum to -6.85 and +2.46 points, and the largest terms are the trimer's $(T_g/300)^n$ (±1.70), $Kr_2^* + F \rightarrow KrF(B) + Kr$ (±0.91), $KrF(B,C) + F \rightarrow Kr + 2F$ (±0.88), $e^- + Kr^* \rightarrow Kr^+ + 2e^-$, pre-exponential ($\pm0.29$). On Tahoma's gain (5.93 %/cm) it is ±33 %, the largest terms the trimer's $(T_g/300)^n$ (±1.55), KrF^* per ion-ion recombination (±0.75), σ_{se}, 248-nm stimulated emission (±0.71), $KrF(B,C) + F \rightarrow Kr + 2F$ (±0.47); the measurement is -7 % away, 0.2 quadrature units. On the absorption (0.82 %/cm) it is ±37 %, the largest terms $Kr_2F^* + F_2$ (±0.18), $Kr_2F^* + F \rightarrow 2Kr + 2F$ (±0.16), $\sigma(Kr_2F^*)$ at 248 nm (±0.12), the trimer's $(T_g/300)^n$ (±0.10); the measurement is -26 % away, 0.9 units. On the net gain (5.11 %/cm) it is ±40 %, the measurement -3 % away, 0.1 units. Every Tahoma measurement lies within one quadrature unit, and the measured cross section has taken the 5s $\leftrightarrow$ 5p term from v0_4's ±2.28-point sum down to its share here; what remains is the trimer's temperature law, on which Section 6 now has a laser to say something.

input	its two ends	248 nm (% of E_{dep})	g_0 (%/cm)	α_0 (%/cm)	$g_0 - \alpha_0$
$Kr_2^* + F \rightarrow KrF(B) + Kr$	0 to 1.0×10^{-9} cm^3 s^{-1}	5.68 to 7.50	5.93 to 6.27	0.82 to 0.82	5.11 to 5.45
$Kr_2^* + Kr_2^* \rightarrow Kr_2^+ + 2Kr + e^-$	0 to 5.0×10^{-10} cm^3 s^{-1}	5.68 to 5.69	5.93 to 5.93	0.82 to 0.82	5.11 to 5.11
$Kr_2F^* + F \rightarrow 2Kr + 2F$	0 to 1.8×10^{-10} cm^3 s^{-1}	5.68 to 5.84	5.93 to 6.07	0.82 to 0.51	5.11 to 5.56
$Kr^* + 2Kr \rightarrow Kr_2^* + Kr$, 300 K	1.7×10^{-32} to 5.5×10^{-32} cm^6 s^{-1}	5.68 to 5.22	5.93 to 5.57	0.82 to 0.82	5.11 to 4.75
$Kr_2^* + F_2$, both channels	2.1×10^{-10} to 5.0×10^{-10} cm^3 s^{-1}	5.68 to 5.70	5.93 to 5.98	0.82 to 0.83	5.11 to 5.15
$Kr^* + Kr^* \rightarrow Kr^+ + Kr + e^-$	1.5×10^{-10} to 5.0×10^{-10} cm^3 s^{-1}	5.87 to 5.68	5.95 to 5.93	0.82 to 0.82	5.13 to 5.11
$Kr_2F^* + F_2$	1.2×10^{-10} to 4.3×10^{-10} cm^3 s^{-1}	5.68 to 5.86	5.93 to 6.12	0.82 to 0.47	5.11 to 5.66
the trimer's $(T_g/300)^n$	-3.32 to -0.5	5.68 to 2.27	5.93 to 2.83	0.82 to 1.03	5.11 to 1.80
KrF(B,C) + F → Kr + 2F	0 to 7.8×10^{-10} cm^3 s^{-1}	5.68 to 3.93	5.93 to 4.99	0.82 to 0.75	5.11 to 4.24
$Kr^{**} + Kr \rightarrow Kr^* + Kr$	3.0×10^{-11} to 1.7×10^{-10} cm^3 s^{-1}	5.71 to 5.68	5.88 to 5.93	0.94 to 0.82	4.94 to 5.11
$e^- + Kr^* \rightarrow e^- + Kr^{**}$, a scale on the measured cross section	0.5 to 1.4	5.42 to 5.75	5.78 to 6.01	0.82 to 0.82	4.96 to 5.19
$e^- + Kr^* \rightarrow Kr^+ + 2e^-$, pre-exponential	0 to 1.5×10^{-7} cm^3 s^{-1}	5.68 to 5.09	5.93 to 5.95	0.82 to 0.82	5.11 to 5.13
$e^- + Kr^{**} \rightarrow Kr^+ + 2e^-$, pre-exponential	0 to 2.0×10^{-7} cm^3 s^{-1}	5.68 to 5.49	5.93 to 5.94	0.82 to 0.82	5.11 to 5.11
$Kr^{**} + F_2$, all channels	0 to 1.3×10^{-9} cm^3 s^{-1}	5.66 to 5.68	5.92 to 5.93	0.82 to 0.82	5.10 to 5.11
KrF* per ion-ion recombination	0.5 to 1	5.57 to 5.68	4.43 to 5.93	0.66 to 0.82	3.76 to 5.11
share of the ion-ion KrF* in C	0 to 0.5	5.68 to 5.65	5.93 to 5.40	0.82 to 0.81	5.11 to 4.59
the wave's electron-ion absorption, a factor	0.9 to 1.1	5.68 to 5.68	5.93 to 5.93	0.82 to 0.82	5.11 to 5.11
$Kr_2^+ + e^- \rightarrow Kr^{**} + Kr$	±10 % (Shiu and Biondi 1977, ±10 %)	5.68 to 5.69	5.94 to 5.92	0.82 to 0.82	5.12 to 5.11
$Kr^+ + 2Kr \rightarrow Kr_2^+ + Kr$	±9 % (Grössl et al. 1981, $(2.2 \pm 0.2) \times 10^{-31}$)	5.68 to 5.69	5.94 to 5.93	0.82 to 0.82	5.12 to 5.10
Kr_2F^* radiative lifetime	±9 % (Xu et al. 1993, 220 ± 20 ns)	5.69 to 5.68	5.93 to 5.93	0.84 to 0.80	5.09 to 5.13
σ_{se}, 248-nm stimulated emission	±12 % (Rokni et al. 1978, $(2.5 \pm 0.3) \times 10^{-16}$)	–	5.22 to 6.64	–	4.40 to 5.82
$\sigma(Kr_2F^*)$ at 248 nm	±20 % (Hakuta et al. 1987, $(1.0 \pm 0.2) \times 10^{-18}$)	–	–	0.70 to 0.94	4.99 to 5.23

Table 3: The propagated uncertainty's inputs, each alone at its two ends, the others at their defaults, and the four outputs at those ends (low end to high end), under v0_2's closure. 17 minutes of runs.

2 What v0_5 changes

The measured cross section. Jung, Stone, Boffard, Anderson and Lin [13] measured the electron-impact excitation cross sections out of the two metastable levels of Kr, $1s_5$ (J = 2) and $1s_3$ (J = 0), into the 10 levels of the $4p^5 5p$ configuration, at 2 to 10 eV for every level and to 300 eV for the two largest (their Tables II to IV; the Letter [12] prints the 3.5-eV column). The totals at 6 eV are 1.07×10^{-14} cm^2 out of $1s_5$ and 1.27×10^{-14} out of $1s_3$, with stated uncertainties of ±35 and ±40 %. LIGHT's Kr* is the whole $4p^5 5s$ manifold (the Biagi group at 9.915 eV) and its Kr** the whole 5p manifold, so one lumped cross section is needed: the statistical-weight mean over the four 5s levels of each level's sum over the ten 5p levels. Nothing is measured out of the two J = 1 resonance levels, which at 1 atm are radiation-trapped and populated with the metastables; they are given the set of the metastable with the same ion core, their thresholds shifted to their own energies. That is the one assumption, and the band k_5s5p brackets it: 0.5 is the metastables alone (the J = 1 levels contributing nothing, half the manifold's weight) and 1.4 the measurement's upper error. The lumped cross section is 1.14×10^{-14} cm^2 at 6 eV (the metastables alone 5.51×10^{-15}, 0.49 of it), linear between the tabulated energies, from 0 at each pair's own threshold, and above the tables on the shape of the largest column, $1s_5 \rightarrow 2p_9$. The electron loses the ledger's gap, 1.385 eV, as in v0_4, so the ledger closes. Under v0_2's closure the row is the Maxwellian table of that cross section and its reverse the Klein-Rosseland table; under the kinetic EEDF the population process carries the cross section itself. Checks: the solver's forward rate for a Maxwellian matches the table to 1.5×10^{-3} and the reverse to 4.9×10^{-4}, and detailed balance over the gap holds to 9.7×10^{-4} (Section 8).

The finding. The measured set confirms the fit it replaces, Kannari's Table I [76], Jacob and Mangano's $A\,T_e^{1/2}\exp(-C/T_e)$ with $A = 9.0\times10^{-7}$, C = 1.51 eV: the lumped rate is ×1.40 the fit's at 1 eV, ×1.37 at 1.16, ×1.23 at 2, ×1.11 at 3 and ×0.91 at 5.8 eV, above it where the laser's electrons sit and below it at the reference point's. So the partition v0_4 found rests on a measurement now and moves little: Table 5 and Table 4. `Params.k76 = "kannari"` keeps the fit as a switch and reproduces v0_4 to four digits.

A TALIF lifetime. The TALIF review of Gazeli et al. [14] gives the natural lifetime of the Kr $5p'[3/2]_2$ calibration level as 34.1 ns; PRAD's statistical mean of the ten measured level lifetimes is 29.0 ns, and Dzierżęga et al. [16] give that level 28.6 ns. No change; the review tabulates no Kr self-quenching, so KPS stays on Gazeli et al.'s Physics of Plasmas measurement [15].

A second laser. Section 6.

What v0_4 added, and this version keeps unchanged: the 5p rung (KPS at 1.7×10^{-10} cm^{3}s^{-1}, PRAD at 29.0 ns, K27 at 1.3×10^{-9} cm^{3}s^{-1}, the stepwise ionizations as bands at 0), KrF(B,C) + F as a band,

the 248-nm optics on Kannari's absorbers with Hakuta et al.'s measured Kr_2F^* cross section, ion-ion recombination on Flannery's turnover with the Langevin limb of the measured mobilities, the walls of a gap and the EM wave with Turnbull et al.'s inverse bremsstrahlung, the 5s $\leftrightarrow$ 5p collisions, the ions' Coulomb collisions and the wave's field term in the Boltzmann equation. v0_4's overview [22] describes and checks them; their numbers here are unchanged from it (the wave's 0.024 J cm^{-3} beside the 0.3 given, the walls' 9.3×10^{-6} %, the field term's atom part to 6.7×10^{-3} and ion part to 1.0×10^{-3}, `ib.py` held by 219 checks).

3 How the numbers were made

Every number in this note is written by `docs/make_overview_v0_5.py` from v0_5 and from `ML/`, or by `docs/make_system_map_v0_5.py` (the reference point under v0_2's closure, its bands and its optics, which the system map draws), and a test fails when those files no longer match the model. v0_4's numbers come from its own generated files, unchanged. ML runs v0_5 when `LIGHT_VERSION` names it. Tahoma runs under the prototype beam drive, a test fixture in `docs/` and not part of the model, in ML's configuration (v0_2's: the default network with every band at its default and n_e free from a seed of 1.0×10^{10} cm^{-3}). Kleikamp et al.'s laser runs under the same drive at their pump through `docs/krf_data/kleikamp_v0_5.py`. The kinetic EEDF takes minutes a run, so Tahoma's variants, its saturation intensity, its formation efficiency and the Kleikamp scans run on v0_2's closure (the kinetic closure moves v0_5's Tahoma numbers by about 1 %, Table 1); the kinetic numbers are the reference point and Tahoma's default, 19 and 20 minutes each.

4 Tahoma and the optics

v0_2's closure	g_0 (%/cm)	α_0 (%/cm)	$g_0 - \alpha_0$	T_e (eV)	n_e (cm^{-3})
measured [5]	6.39	1.11	5.28		
every band at its default	5.93 ($\times$0.93)	0.82 ($\times$0.74)	5.11 ($\times$0.97)	1.10	4.19×10^{14}
the rung's other bands at Kannari's values (`-ladder kannari`)	5.94 ($\times$0.93)	0.84 ($\times$0.76)	5.09 ($\times$0.96)	1.10	4.28×10^{14}
Kannari's fit in K76's place (`-k76 kannari`, v0_4)	5.86 ($\times$0.92)	0.82 ($\times$0.74)	5.04 ($\times$0.95)	1.18	4.51×10^{14}
the 5s $\leftrightarrow$ 5p cycle off (`k_5s5p` = 0)	5.69 ($\times$0.89)	0.84 ($\times$0.76)	4.84 ($\times$0.92)	1.60	6.32×10^{14}
the measured cross section at the band's low end (0.5)	5.78 ($\times$0.90)	0.82 ($\times$0.74)	4.96 ($\times$0.94)	1.28	4.89×10^{14}
the measured cross section at its high end (1.4)	6.01 ($\times$0.94)	0.82 ($\times$0.74)	5.19 ($\times$0.98)	1.01	3.85×10^{14}
Kr^{**} + Kr at the band's low end	5.88 ($\times$0.92)	0.94 ($\times$0.85)	4.94 ($\times$0.94)	1.20	4.58×10^{14}
KrF(B,C) + F at Kannari's analogue	4.99 ($\times$0.78)	0.75 ($\times$0.68)	4.24 ($\times$0.80)	1.10	4.18×10^{14}
Kr_2^* + F at the top of its band	6.27 ($\times$0.98)	0.82 ($\times$0.74)	5.45 ($\times$1.03)	1.10	4.17×10^{14}
the trimer on $T_g^{-1/2}$	2.83 ($\times$0.44)	1.03 ($\times$0.93)	1.80 ($\times$0.34)	1.10	4.17×10^{14}
ion-ion KrF^* yield at 0.5	4.43 ($\times$0.69)	0.66 ($\times$0.59)	3.76 ($\times$0.71)	1.10	4.17×10^{14}
ion-ion C share at 0.5	5.40 ($\times$0.85)	0.81 ($\times$0.73)	4.59 ($\times$0.87)	1.09	4.2×10^{14}
v0_4	5.86 ($\times$0.92)	0.82 ($\times$0.74)	5.04 ($\times$0.95)	1.18	4.51×10^{14}
v0_3	5.68 ($\times$0.89)	0.84 ($\times$0.75)	4.85 ($\times$0.92)	1.60	6.33×10^{14}
v0_5, kinetic EEDF	5.99 ($\times$0.94)	0.83 ($\times$0.75)	5.16 ($\times$0.98)	1.08	4.07×10^{14}
v0_4, kinetic EEDF	5.91 ($\times$0.93)	0.82 ($\times$0.74)	5.09 ($\times$0.96)	1.16	4.43×10^{14}

Table 4: Tahoma over the 200 to 400 ns window: v0_5 with every band at its default and each variant alone, v0_4 and v0_3 (v0_4's cross sections on their densities), and the kinetic EEDF. The absorption by species with the defaults: Kr_2F^* 0.59, F^- 0.10, F_2 0.06, Kr^{**} 0.03, Kr_2^+ 0.03, Kr^* 0.01 %/cm (v0_2's closure), Kr_2F^* 0.60, F^- 0.10, F_2 0.06, Kr^{**} 0.03, Kr_2^+ 0.03, Kr^* 0.01 (kinetic); pure Kr, no F_2: Kr^* 0.11, Kr^{**} 0.10, Kr_2^+ 0.04.

At Tahoma's 1.10-eV electrons the measured cycle is $\times$1.37 the fit's, cools the electrons a little further (v0_4's 1.18 eV) and lifts the gain by +1 %; the absorption does not move, 72 % of it being Kr_2F^*. The band's two ends, the metastables alone and the measurement's upper error, move the gain between

$\times 0.90$ and $\times 0.94$; the cycle off gives $\times 0.89$. The bands that move the gain most are v0_4's: the trimer's temperature law ($\times 0.44$), the ion-ion yield ($\times 0.69$), KrF(B,C) + F at Kannari's analogue ($\times 0.78$), the ion-ion C share ($\times 0.85$) and Kr_2^* + F ($\times 0.98$), none with a measurement behind the band. In the window under the kinetic EEDF LIGHT's mean energy is the kinetic one at every update, held by a field of at most 0.11 Td, and the two energy balances agree to 4.3×10^{-4} of the beam's electron power. ML's formation run gives g_0 5.93 %/cm with its own channels, I_{sat} 1.45 MW/cm^2 and $g_0 I_{sat}/P_{in}$ = 0.261.

5 The reference point

At the notes' reference point, 1 atm, 0.5 % F_2, 300 kW cm^{-3} for 1 μs and run to 2 μs, v0_5 gives these shares of the deposited energy:

	v0_5, kinetic	v0_5, v0_2's closure	v0_4, kinetic	v0_4, v0_2's closure
248 nm, KrF B-X	5.73	5.7	5.68	5.79
272 nm, KrF B-A	0.183	0.2	0.181	0.185
275 nm, KrF C-A	0.829	0.8	0.825	0.821
420 nm, Kr_2F^*	4.21	3.9	4.28	3.84
147 nm, Kr_2^*	4.8	4.6	5.3	4.5
near IR, 5p $\rightarrow$ 5s	0.28	0.3	0.27	0.28
all bands	16.1	15.4	16.5	15.5
heat, reactions	72.6	71.6	71.9	71.6
to Kr^*, gross	64.4	60.7	65.4	60.6
to Kr^+	3.6	3.2	3.9	3.2
KrF(B) formed	24.2	23.0	24.3	23.0
elastic heating	1.8	3.4	2.0	3.4
g_0, 500 to 1000 ns (%/cm)	3.79	3.96	3.73	4.12
α (%/cm)	0.67	0.64	0.68	0.63
T_g at the end (K)	738	742	735	742
T_e, pulse mean (eV)	2.96	5.68	2.97	5.81

Table 5: The reference point, percent of E_{dep} unless a row says otherwise. v0_5's v0_2-closure column is the system map's run; v0_4's is the same run with Kannari's fit in K76's place.

What moves the result is still the 5s $\leftrightarrow$ 5p cycle, on the measurement now: the electrons pump 80.9 % of the Kr^* formed up to the rung (79.4 % under the kinetic EEDF; v0_4 78.4 %), the rung relaxes 96.0 % of it on Kr and the 1.385 eV of every cycle heats the gas, 33.7 % of the input. Under v0_2's closure the electrons sit at 5.68 eV, where the measured rate is $\times 0.92$ the fit's, so the share falls from v0_4's 5.79 to 5.7 % and the Kr_2^* route gains it; under the kinetic EEDF they sit at 2.96 eV, where it is $\times 1.11$, and the share rises from 5.68 to 5.73 %. With the rung neutralised the share is 4.67 %, v0_3's. Under the kinetic EEDF the field carries 1.001 to 1.003 of the input through the pulse's second half and the two energy balances agree to 6.6×10^{-6}; the run takes 425 kinetic solutions, 277 in the pulse, with E/N between 1.56 and 45 Td, and closes its ledger below 10^{-14}.

Every band's two ends under v0_2's closure, one band at a time (`run_v0_5.py -bands`):

band	default	range	248 nm	147 nm	g_0	α	T_g (K)
the trimer's $(T_g/300)^n$	-3.32	-3.32 to -0.50	5.68 to 2.27	4.6 to 4.3	3.96 to 1.21	0.64 to 0.96	742 to 754
KrF(B,C) + F → Kr + 2F	0	0 to 7.8×10^{-10} cm^3 s^{-1}	5.68 to 3.93	4.6 to 4.4	3.96 to 2.32	0.64 to 0.49	742 to 761
Kr_2^* + F → KrF(B) + Kr	0	0 to 1.0×10^{-9} cm^3 s^{-1}	5.68 to 7.50	4.6 to 0.2	3.96 to 5.73	0.64 to 0.65	742 to 757
e^- + Kr* → e^- + Kr**, a scale on the measured cross section	1	0.5 to 1.4	5.42 to 5.75	7.2 to 3.5	3.59 to 4.10	0.70 to 0.60	724 to 750
e^- + Kr* → Kr^+ + 2e^-, pre-exponential	0	0 to 1.5×10^{-7} cm^3 s^{-1}	5.68 to 5.09	4.6 to 2.1	3.96 to 3.70	0.64 to 0.54	742 to 770
Kr* + 2Kr → Kr_2^* + Kr, 300 K	1.7×10^{-32}	1.7×10^{-32} to 5.5×10^{-32} cm^6 s^{-1}	5.68 to 5.22	4.6 to 11.0	3.96 to 3.66	0.64 to 0.69	742 to 704
e^- + Kr** → Kr^+ + 2e^-, pre-exponential	0	0 to 2.0×10^{-7} cm^3 s^{-1}	5.68 to 5.49	4.6 to 3.7	3.96 to 3.84	0.64 to 0.60	742 to 751
Kr* + Kr* → Kr^+ + Kr + e^-	5.0×10^{-10}	1.5×10^{-10} to 5.0×10^{-10} cm^3 s^{-1}	5.87 to 5.68	6.0 to 4.6	4.12 to 3.96	0.67 to 0.64	730 to 742
Kr_2F^* + F_2	1.2×10^{-10}	1.2×10^{-10} to 4.3×10^{-10} cm^3 s^{-1}	5.68 to 5.86	4.6 to 4.5	3.96 to 4.06	0.64 to 0.39	742 to 754
Kr_2F^* + F → 2Kr + 2F	0	0 to 1.8×10^{-10} cm^3 s^{-1}	5.68 to 5.84	4.6 to 4.5	3.96 to 4.07	0.64 to 0.26	742 to 757
KrF* per ion-ion recombination	1	0.5 to 1	5.57 to 5.68	4.6 to 4.6	3.88 to 3.96	0.63 to 0.64	743 to 742
share of the ion-ion KrF* in C	0	0 to 0.5	5.68 to 5.65	4.6 to 4.6	3.96 to 3.94	0.64 to 0.64	742 to 742
Kr** + Kr → Kr* + Kr	1.7×10^{-10}	3.0×10^{-11} to 1.7×10^{-10} cm^3 s^{-1}	5.71 to 5.68	5.5 to 4.6	3.96 to 3.96	1.07 to 0.64	727 to 742
Kr** + F_2, all channels	1.3×10^{-9}	0 to 1.3×10^{-9} cm^3 s^{-1}	5.66 to 5.68	4.5 to 4.6	3.93 to 3.96	0.63 to 0.64	743 to 742
Kr_2^* + F_2, both channels	2.1×10^{-10}	2.1×10^{-10} to 5.0×10^{-10} cm^3 s^{-1}	5.68 to 5.70	4.6 to 3.3	3.96 to 3.93	0.64 to 0.66	742 to 747
Kr_2^* + Kr_2^* → Kr_2^+ + 2Kr + e^-	0	0 to 5.0×10^{-10} cm^3 s^{-1}	5.68 to 5.69	4.6 to 3.4	3.96 to 3.94	0.64 to 0.63	742 to 749
the wave's electron-ion absorption, a factor	1	0.9 to 1.1	5.68 to 5.68	4.6 to 4.6	3.96 to 3.96	0.64 to 0.64	742 to 742

Table 6: Each band alone at its two ends, the others at their defaults (17 bands, 7 with no measurement and a default of 0). The columns are the 248-nm and 147-nm shares of E_{dep} (%), the 500-to-1000-ns gain and absorption (%/cm) and the final gas temperature, low end to high end. With every band at its default the 248-nm share is 5.7 %.

6 A second laser, held out: Kleikamp et al.'s Kr/F_2 laser

Kleikamp, Peters and Witteman [17] pumped a KrF laser with krypton as the only buffer in a coaxial e-beam diode (280 kV, 250 A/cm^2 on average, 30-ns pulses, electrons of at most 170 keV in the gas, a 1-cm tube 20 cm long, an outcoupling of 5.75×10^{-2} cm^{-1}). With F_2 at 10 mbar the output peaks at about 2 atm of Kr (their Fig. 3); with Kr at 2 atm at about 10 mbar of F_2 (Fig. 4); and their steady-state cavity balance deduces a three-body KrF* + Kr quenching constant of 4.6×10^{-31} cm^{6}s^{-1} (their Table I: "the quenching constants of Kr and Ne are not known in the literature"). It is a held-out device, a score and never a target.

LIGHT runs under the beam drive at their conditions. The deposited power is not printed in the paper: from the collision stopping power of Kr at 170 keV in Berger and Seltzer's tables [18], the source the paper names, read to ±15 %, it is 3.8 MW cm^{-3} at 2 atm; from their own pumping rate B_{Kr} with the W-value of Kr, 5.6. The scans run at the mean, 4.7 MW cm^{-3} at 2 atm proportional to the density, and again at 0.7 and 1.4 times it. LIGHT has no cavity, so two proxies for the output are compared with the measured optimum's location over the pulse's second half (15 to 30 ns): the net small-signal gain $g_0 - \alpha_0$, and the paper's own balance of a saturated laser (its Eqs. (1) to (3)): with the inversion clamped at threshold, $[\text{KrF}]_{\text{thr}} = (\gamma + \alpha)/\sigma_s$, the output per length is $\gamma/(\gamma + \alpha)\,(P - Q[\text{KrF}]_{\text{thr}})$ with P LIGHT's KrF(B) formation rate, Q its loss rate per molecule (quenching and radiation) and α its absorption. The optimum of the second balances the pumping, which rises with the density, against the quenching of the threshold inversion, which rises faster (the trimer goes as the density squared), so it tests the trimer's rate at the laser's gas temperature, the largest term of the propagated uncertainty.

Kr (atm)	g_0 (%/cm)	α_0	$g_0 - \alpha_0$	Q (s^{-1})	output, saturated balance	T_g (K)
0.25	14.4	0.63	13.8	3.77×10^8	0.24	352
0.5	20.4	0.83	19.6	5.72×10^8	0.60	357
0.75	21.1	1.14	19.9	8.38×10^8	0.85	359
1	19.8	1.50	18.3	1.18×10^9	1.00	361
1.5	16.0	2.27	13.7	2.08×10^9	1.00	362
2	12.7	3.02	9.7	3.29×10^9	0.70	362
2.5	10.2	3.72	6.4	4.81×10^9	0.15	361
3	8.2	4.37	3.9	6.67×10^9	-0.62	361
3.5	6.8	4.96	1.8	8.86×10^9	-1.60	360

Table 7: LIGHT v0_5 at Kleikamp et al.'s conditions, F_2 at 10 mbar, the nominal pump, v0_2's closure, the 15 to 30-ns window. The saturated-balance output is relative to its maximum. The measured output peaks at about 2 atm.

The saturated balance peaks at 1.26 atm (1.02 at 0.7 times the pump, 1.48 at 1.4), the small-signal net gain at 0.67; the laser peaked at 2. At 2 atm LIGHT's KrF(B) loss rate is $3.29\times10^{9}\,s^{-1}$ at the window's T_g of 362 K (T_e 1.23 eV, n_e $1.5\times10^{15}\,cm^{-3}$), the trimer most of it at $1.02\times10^{-30}\,cm^{6}s^{-1}$ (G5's 300-K rate on the $(T_g/300)^{-3.32}$ law), $\times2.2$ the constant the paper's balance deduces. With G5 and G5C scaled by 0.45, which brings the effective quenching to 4.57×10^{-31}, the saturated optimum moves to 1.83 atm (the small-signal one to 0.99); that is a sensitivity run and not adopted, the laser's constant being a model-dependent number (their Q is three-body only, their output insensitive to the outcoupling) that bounds the term rather than fixes it. The F_2 optimum is not reproduced: both proxies still rise at 30 mbar where the laser peaked at 10, because LIGHT's absorption falls with F_2 (Kr_2F^* is quenched away faster than F_2 and F^- absorb). That points at the F_2-side absorbers, F_2's compilation value and F^-'s 1971 measurement, and at KrF^* + F_2 quenching, none of them moved here. A proxy for laser output must saturate: the small-signal net gain quenches the whole inversion and peaks a factor two low in pressure; the paper's balance is the honest proxy, and even it falls short of the measurement.

7 The energy ladder

v0_5's ladder (Figure 1) is v0_4's with K76 and K76r on the measurement: the 5p rung 11.3 eV above Kr and 1.39 eV above Kr*, KPS and K27X as quenches, PRAD's 1.39-eV photon, the two stepwise ionizations and the harpoon from 5p; Kr_2^+ + e^- lands on it, leaving 1.55 eV to the gas. KrF(B) lies 5.54 eV below the ion pair Kr^+ + F^- and 5.06 eV above Kr + F, and the 248-nm photon carries 5.00 eV of that.

8 The new terms' checks

- The tables as printed: spot values of Jung et al.'s Tables II to IV are pinned (`test_v0_5.py`), and the sums at 6 eV are 1.07×10^{-14} and $1.27\times10^{-14}\,cm^2$.
- The Maxwellian table of the lumped cross section against a direct 120 001-point trapezoid at 1, 2 and 5.8 eV: 2×10^{-3}.
- The solver's population process on 2400 cells to 40 eV: the forward rate for a Maxwellian against the table to 1.5×10^{-3}, the reverse against the Klein-Rosseland table to 4.9×10^{-4}, detailed balance over the gap to 9.7×10^{-4}.
- The switch: `k76 = "kannari"` reproduces v0_4's reference point (Table 5's last column against v0_4's overview) to four digits.
- A 60-ns kinetic pulse with the measured pair inside the equation: the field carries 1.003 to 1.007 of the input and the balances agree to 2.0×10^{-7}.
- v0_4's checks, unchanged: the wave's field term (atoms to 6.7×10^{-3}, the marginal ion part to 1.0×10^{-3}), the ion-ion limbs (2.61×10^{-6} and $2.74\times10^{-6}\,cm^3s^{-1}$ at 1 amagat), the walls' $61\,s^{-1}$ at mid-pulse, `ib.py` by 219 checks.

9 Where v0_5 is not yet reliable

- The 5s $\rightarrow$ 5p excitation is measured; the two J = 1 levels are not, and their proxies are the band's low end. Jung et al.'s absolute scale is $\pm$35 to 40 %, the band's high end.
- The trimer's rate at the laser's gas temperature: the one laser that bears on it finds LIGHT's quenching $\times2.2$ too strong at 362 K, through a model-dependent constant. The temperature law is still the largest term of the propagated uncertainty.
- The F_2-side absorbers and KrF^* + F_2: the laser's F_2 optimum is not reproduced.
- KPS one level's total loss, K27 a search snippet, the stepwise ionizations and the 5p harpoon as LIGHT-side rates, the absorbers as a diagnostic, the wall closure, no propagation: as in v0_4.

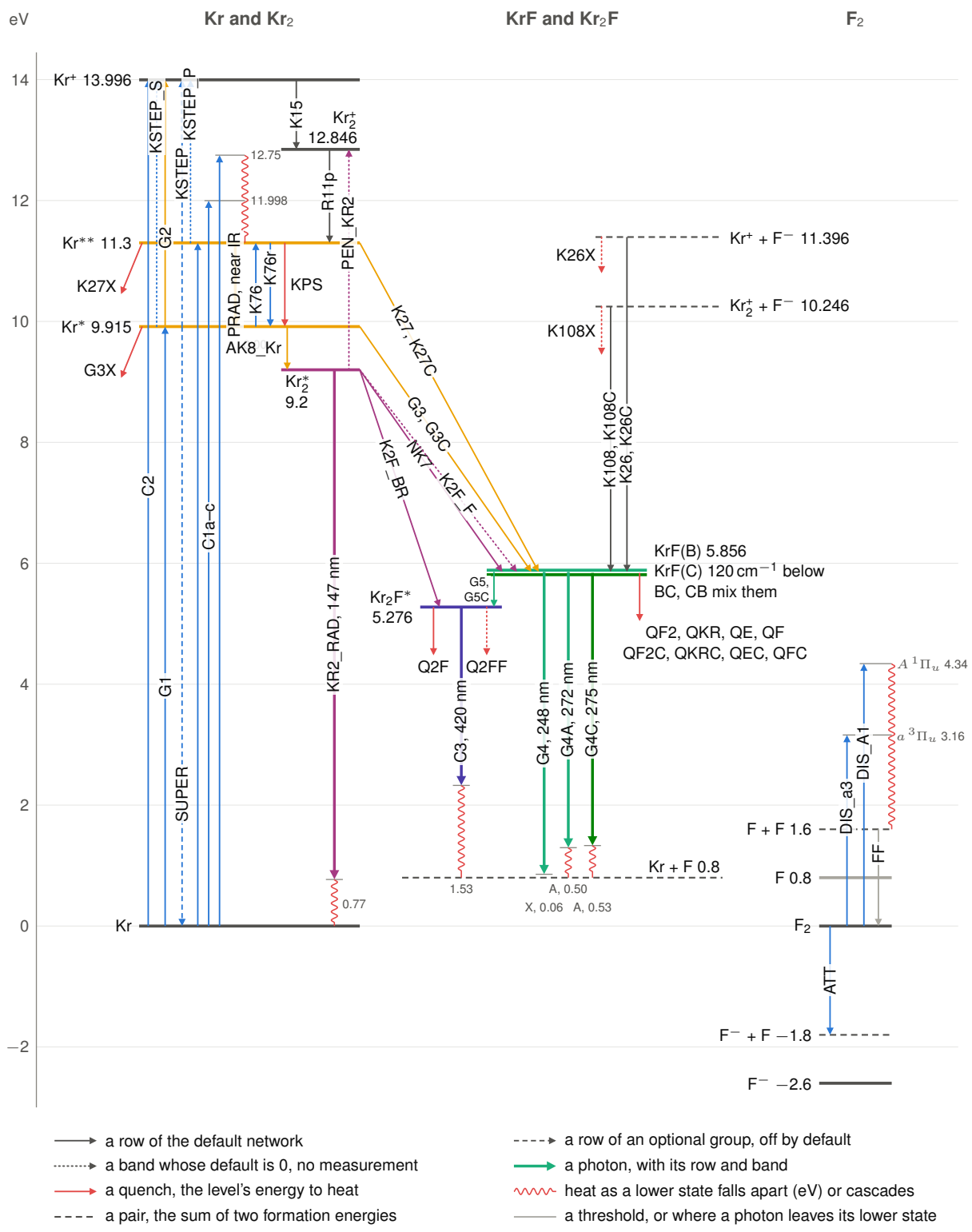


Figure 1: The energy ladder of LIGHT v0_5, version 0.1.5, in eV above ground-state Kr and F_2 at rest. A bar is a species at its formation energy and a dashed bar a pair, the sum of two. An arrow is a row, labelled with its id. It is solid in the default network and dotted for a band whose default is 0, Q2FF, K26C, K26X, K108C, K108X, K2F_F, PEN_KR2, KSTEP_S, KSTEP_P, QF and QFC. The rows of the optional group `electron` are dashed. A photon's arrow drops by its quantum, and the red squiggle below it is the heat its lower state gives the gas as it falls apart. A red stub is a quench. The energy a row releases heats the gas, except a photon's, which leaves, and that of G2, PEN_KR2, K76r and SUPER, which goes to the electrons. KrF(B) and KrF(C) lie 120 cm^{-1} apart and are drawn as a pair of bars that every B and C row meets. The rows G6, VIB1, VIB2, VIB3 and VIB4 move nothing between levels and are not drawn.

- Everything in v0_3's list: the steady EEDF after the pulse, the first cell, the thermal electrons' grid bias, Kr_2^* + F, the trimer above its measured range, Hakuta's and Burnham and Searles' conflicts, and the pure-krypton VUV yield at low pump.

A Reaction rates

v0_5's rows are v0_4's (its overview's Appendix A, and v0_2's) with these two changed. Section 8 of `v0_5/SOURCES.md` says how each source was read.

row	reaction	value	source
K76	$e^- + Kr^* \rightarrow e^- + Kr^{**}$	the lumped measured cross section, 1.14×10^{-14} cm^2 at 6 eV; $k = 3.61\times10^{-7}$ at 1.16 and 1.52×10^{-6} cm^{3}s^{-1} at 5.8 eV (Maxwellian); band k_5s5p a scale 0.5 to 1.4	Jung et al. [13] Tables I to IV, [12] Table I; the $J = 1$ levels on their same-core metastable's set; Kannari's fit [10, 11] as `k76 = "kannari"`
K76r	$e^- + Kr^{**} \rightarrow e^- + Kr^*$	the Klein-Rosseland reverse, g 12/36 over the gap	detailed balance
PRAD (unchanged)	$Kr^{**} \rightarrow Kr^* + h\nu$	τ = 29.0 ns	Dzierżęga et al. [16]; the TALIF review's 34.1 ns for $5p'[3/2]_2$ [14] as a cross-check

Table 8: The rows v0_5 changes. Rates in cm^3/s.

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